A Place-Based Intelligent Learning Technology Leveraging Microclimates for Adaptation

Imran S. Afizullah-Khan, Emmanuel G. Blanchard(✉), and Sébastien George

Laboratoire d'Informatique de l'Université du Mans, Le Mans University, Le Mans, France
{imran.khan,emmanuel.blanchard,sebastien.george}@univ-lemans.fr

Abstract. Climate education is an urgent global priority, yet many existing approaches remain abstract and disconnected from learners' local experiences. This study explores the potential of microclimates within local settings like university campuses, city squares, and local nature parks as an observable context for place-based climate education. Addressing the limited research on leveraging microclimatic conditions for instructional purposes, we propose an approach that integrates Internet of Things (IoT) sensors-based microclimatic data such as ambient temperature, humidity and light, with Generative Artificial Intelligence to create contextualized and adaptive learning experiences. A pilot study with university students yielded promising outcomes in knowledge gain, motivation, and system acceptance, contributing novel insights to the emerging field of AIED for Place-based climate education.

Keywords: Climate Education · Place-based Education · Adaptive Learning · Generative AI in Education · Education Chatbot

1 Introduction and Prior Work

Climate education is widely recognized as an urgent global priority. Reports from international organizations like OECD [9] emphasize the far-reaching impacts of climate change on daily life, public health, economic and environmental sustainability. Existing approaches often emphasize abstract explanations and global phenomena while neglecting how climate change is experienced in learners' local context [9]. Place-based learning [13] is an approach that situates learning within learners' local environments and real-world contexts to enhance relevance and understanding. In this regard, places such as university campuses, nature parks, and even urban settings offer significant potential for place-based climate education. While this education approach has been explored through outdoor summer programs, classroom-based local climate instruction, and place-based film-making projects [3].

In this study, we adopt the concept of microclimates [7] to make climate change more tangible and locally observable, thereby supporting a clearer and more meaningful understanding. A microclimate refers to localized climate conditions that differ from the surrounding environment. For example, a shaded park may be cooler and more humid

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than a nearby parking lot. On university campuses, city centers and nature parks, microclimates can emerge due to variations in vegetation, building density, surface materials, and human activity, influencing factors such as temperature, humidity, wind flow, and solar radiation. These small-scale variations provide a compelling learning source for climate education, especially in place-based learning scenarios. For example, many people notice changes such as earlier springs, hotter summers, and prolonged rainfall; they often lack an understanding of the localized mechanisms driving these phenomena and the role microclimates play in shaping them. By grounding climate education in places familiar to learners such as university campuses, local nature parks and their associated microclimates, learning can move beyond broad generalizations, fostering deeper understanding and encouraging more actionable responses.

In educational settings, microclimates have been examined for their role in enhancing learning environments and influencing learners' cognitive and affective states [8, 11]. However, their use as an instructional context for place-based climate education remains underexplored, likely due to the complexity of implementation through traditional methods. Recent advances in Generative Artificial Intelligence (GenAI) and the Internet of Things (IoT) [12] offer new opportunities to integrate and adapt real-time microclimate data into learning experiences. To this end we try to answer the following research questions in this study:

RQ1. Does a GenAI-based learning system have a measurable impact on climate education outcomes?
RQ2. Does a microclimate-aware learning system produce greater learning impact compared to a non microclimate-aware system?
RQ3. Does a microclimate-aware learning system enhance learner motivation compared to a non microclimate-aware system?

2 Microclimate-Adaptive Place-Based Learning Technology

This section describes how the microclimate aware learning system is implemented as an educational chatbot, covering its theoretical foundations and technical architecture, followed by illustrations of the system in use.

2.1 Theoretical Background

The system is built on four design principles: i) Domain-guided content generation grounds responses in structured climate knowledge and a human-made learning plan; ii) Contextualized instruction adapts climate concepts to real-time microclimatic conditions, replacing abstract explanations with observable phenomena; iii) Instructional scaffolding introduces concepts step by step with guiding questions to support progressive understanding; and iv) the AIED Unplugged [2] framework prioritizes lightweight, accessible learning via Small Language Models (SLM) [6] running on a Jetson Orin Nano Super microcomputer [4] enabling place-based education without cloud dependency. The principles are implemented through context engineering [1], combined with Retrieval Augmented Generation (RAG) [5] to structure microclimate data inputs, domain knowledge, and instructional constraints before passing them to the language model. This approach functions as a control layer between the learner and the AI system, enabling educators to influence how learning experiences are shaped and delivered.

2.2 System Architecture and Design

Figure 1a presents the system architecture and conceptual design, while Figs. 1b and 1c illustrate example dialogues from the implemented system across different scenarios. As shown in Fig. 1a, the user interacts with the client application (i), which triggers the collection of microclimate data including temperature, humidity, and light from the local campus (ii) via Atmo-Station, a microcontroller-based sensor (ESP32) device that captures real-time environmental conditions. The microclimate data together with user response is then forwarded to the context generator, a component responsible for constructing prompts by integrating microclimate data with relevant learning topics and pedagogical rules (iii, iv). Based on the received information, the context generator then retrieves instructional materials from the Knowledge component, a repository of learning plans, and educational content stored as documents or databases (v) and generates a prompt. This prompt is forwarded to the Language Model, which then generates appropriate learning content (vi), and finally, the output is sent to the client (vii).

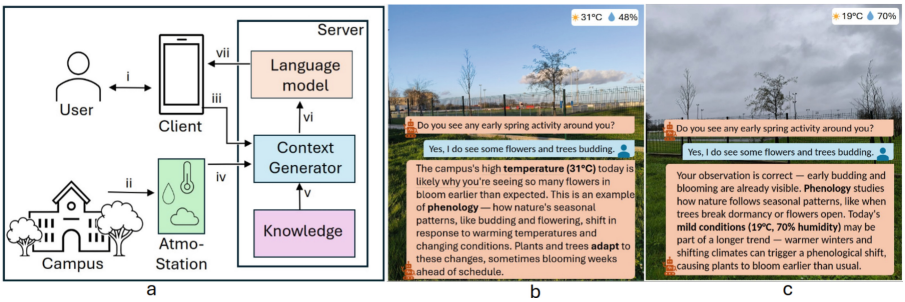


Fig. 1. (a) System architecture and conceptual design. (b) Screenshot under warm and bright conditions. (c) Screenshot under cold and cloudy conditions.

Figure 1b depicts a chatbot interaction under warm, sunny conditions (31 °C, 48% humidity), while Fig. 1c shows the same chatbot operating under cool, cloudy conditions (19 °C, 70% humidity). In both cases, the system delivers instruction on the topic of "phenology" (i.e. the science of recurring biological events in nature and how they are influenced by environmental conditions) but dynamically adapts its examples to reflect the live microclimate data captured at the time of interaction, making the contrast in adaptive responses clearly visible between different conditions.

3 Methodology

A total of 30 university students were recruited for this study, comprising 10 female and 20 male participants (mean age = 25.3, SD = 3.34), drawn from diverse academic backgrounds including science, humanities, engineering, and applied sciences. Given the "pilot" nature of this Design-Based Research study, a small sample was deemed appropriate to make initial assessments of feasibility, usability, knowledge gain, motivation, and user feedback. Two conditions were compared: a test condition delivering learning

activities adapted to micro-climate data collected in real time on a university campus, and a control condition covering the same topics with no micro-climate adaptation. The core learning objective was students' understanding of climate concept of phenological shifts and their underlying causes, including urban heat islands, the albedo effect, and evapotranspiration.

The study was conducted outdoors on the university campus. Participants completed a demographics and consent form, followed by a 6-items multiple-choice pre-test designed to cover all core learning topics mentioned above and establish baseline knowledge. They were then randomly assigned to either the test ($n = 15$) or control ($n = 15$) chatbot for the learning task. A post-test consisting of six items, identical to those used in the pre-test, was administered, followed by a 10-item version of the Intrinsic Motivation Inventory (IMI) measured on a 5 point Likert scale, covering the subscales of interest/enjoyment, perceived competence, value, effort, and pressure/tension. Additionally, three open-ended questions were included to gather participants' opinions on topics such as the use of generative AI for climate education, their most favourite feature, other domains where such technology could be beneficial, and their overall experience. A modified version of the IMI was employed to reduce participant burden and align the instrument with the specific context of the study, an approach consistent with prior research by Pesonen et al. [10] that has adapted selected IMI subscales to suit particular experimental settings. Pre- and post-test scores were analyzed using paired-samples t-tests to assess within-group learning gains, and further compared between the control and experimental conditions. IMI responses were analyzed using the Mann–Whitney U test to examine differences in motivation-related measures between the two groups.

4 Results and Discussions

4.1 General Knowledge Gain

To address RQ1 (i.e., Whether a GenAI-based learning system has a measurable impact on climate education outcomes), pre- and post-test scores were examined using paired-samples t-tests within each condition as mentioned in the methodology section. Participants in the experimental condition exhibited statistically significant increase in knowledge gain following the intervention (pre: $M = 2.33$, $SD = 1.45$; post: $M = 4.13$, $SD = 1.64$; $t(14) = 5.08$, $p < .001$). Notably, participants in the control condition also showed a statistically significant improvement (pre: $M = 2.13$, $SD = 1.36$; post: $M = 4.00$, $SD = 1.73$; $t(14) = 5.14$, $p < .001$). These findings indicate that our system can contribute to measurable gains in climate education outcomes, whether micro-climate adaptation is activated or not.

4.2 Knowledge Gain with Microclimate-Adaptive System

To address RQ2 (i.e., Whether a microclimate-aware learning system produces greater learning impact compared to a non-microclimate-aware system), an independent-samples t-test was conducted as mentioned in the methodology section. Results showed no statistically significant difference in post-test scores between the test and control conditions ($t(28) = 0.22$, $p = .830$), nor in knowledge gain (experimental: $M = 1.80$, $SD =$

1.37; control: M = 1.87, SD = 1.41; t(28) = 0.13, p = .896). These findings suggest that, for the current sample, there is no clear evidence that the microclimate-aware feature produced additional gains, with both systems supporting comparable improvements in learning outcomes. This lack of a between-group difference may be due, in part, to the study being underpowered given the relatively small sample size (n = 30). Additionally, the limited number of assessment items (six questions in both the pre- and post-tests) may have reduced the sensitivity of the measurement. We deliberately chose a compact 6-item pre- and post-test because the activity was conducted on a voluntary basis under real-world conditions, and we wanted to avoid discouraging potential participants with a longer evaluation.

4.3 Perceived Motivation

To address RQ3 (i.e., Whether a microclimate-aware learning system enhances learner motivation compared to a non-microclimate-aware system), the IMI items were analyzed using Mann–Whitney U tests as mentioned in the methodology section. Participants in the test condition reported statistically significantly higher levels of enjoyment than those in the control group (M = 4.47 vs. 3.93; U = 63.50, p = .029). They also rated the test activity as more interesting (M = 4.47 vs. 4.00; U = 66.00, p = .037) and perceived it as easier to understand (M = 4.47 vs. 3.93; U = 63.00, p = .019). These initial results suggest that microclimate adaptive learning content could effectively enhance learners' intrinsic motivation and perceived ease of understanding. Suggesting, aligning learning activities with microclimatic conditions may enhance cognitive performance and motivation, representing a promising avenue for future AIED research. Finally, other IMI items yielded non statistically significant results between the test and control conditions.

4.4 Qualitative Analysis

Participants shared their views on topics such as using GenAI for climate education, favorite app features, and potential additional use cases via open-ended questions. Out of 30, 28 participants found the activity engaging, accessible, and effective at making complex climate concepts more understandable, with the interactive Q&A format, structured AI feedback, and microclimate adaptive content cited as the most valued features. 27 out of 30 participants noted that the conversational nature of the tool made it particularly suitable for learners with limited prior knowledge, and many expressed willingness to use it again or recommended extending it to other educational domains such as history, civic education, and languages. Moreover, 7 participants' perceptions were influenced by a lack of prior information about the system's ethical and AIED unplugged design, leading to concerns about the environmental cost of GenAI and the perceived irony of using an energy-intensive tool for climate education, misconceptions that could have been reduced with clearer communication. 2 participants noted that interactions occasionally felt robotic or impersonal, particularly when addressed in the third person, likely reflecting limitations of the SLM.

5 Conclusion

This paper presents a GenAI and IoT-powered framework for microclimate-adaptive place-based learning, specifically designed for low-resource and autonomous settings. The findings indicate that while the GenAI-based system effectively supports learning overall, the added value of microclimate-aware adaptation in terms of knowledge gain remains inconclusive. Despite this, the system showed promising gains in motivation, such as increased interest and perceived ease of understanding, which may support deeper cognitive learning and merit further investigation. Accordingly, future work will involve a large-scale evaluation incorporating a greater number of pre- and post-test items, guided by statistical power analysis, to improve the sensitivity of the measurement and better detect potential differences between conditions. Furthermore, the research will also focus on exploring methods for refining conversational tone and UI, while also emphasizing transparency to enhance user satisfaction and learning outcomes. Nevertheless, to the best of our knowledge, this study is among the first to explore GenAI for place-based adaptive learning, offering novel insights into this emerging area. Its frugal design indicates high potential for deployment in expert-scarce and low-resource environments. Finally, for the AIED community, it provides a practical blueprint for developing microclimate-adaptive, sustainable, and human-centered GenAI-powered learning systems, opening new directions for research at the intersection of GenAI, IoT, and adaptive education.

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